ABSTRACT

Air pollution has become a major environmental and public health concern, particularly in densely populated urban areas. The concentration of air pollutants can change significantly over time due to variations in weather conditions, human activities, and environmental factors. Conventional air quality monitoring systems primarily provide information about existing pollution levels, offering limited insight into future changes. Therefore, predicting air quality in advance is essential for supporting timely decisions and improving pollution awareness.

This project presents **AirCast: Intelligent Air Quality Prediction System**, a machine learning-based system designed to predict the next-hour concentration of fine particulate matter (PM2.5). The system utilizes the Beijing Multi-Site Air Quality Dataset, which contains hourly air quality and meteorological observations collected from 12 monitoring stations in Beijing between 2013 and 2017. The dataset includes pollutant measurements such as PM2.5, PM10, SO2, NO2, CO, and O3, along with meteorological parameters such as temperature, pressure, dew point, rainfall, wind direction, and wind speed. After data preparation and target generation, 412,017 records were retained for the modelling process.

The methodology involves data cleaning, handling missing values, feature engineering, temporal feature extraction, and the generation of historical PM2.5 lag features. A chronological data-splitting strategy is adopted to preserve the temporal order of observations and prevent information leakage. Multiple machine learning regression algorithms, including Random Forest, Extra Trees, XGBoost, CatBoost, LightGBM, and HistGradientBoosting, are evaluated and compared. Based on the validation results, a tuned Random Forest Regressor is selected as the final prediction model.

The selected model achieves a Mean Absolute Error (MAE) of 10.74, Root Mean Squared Error (RMSE) of 21.36, and coefficient of determination (R^2) of 0.9533 on the held-out test dataset. The proposed system is intended to provide predictive air quality information through a Flask-based web application, helping users understand potential changes in pollution levels and make more informed decisions.